

- 5 A metal ball is attached to the lower end of a spring, as shown in Figure 5.1.

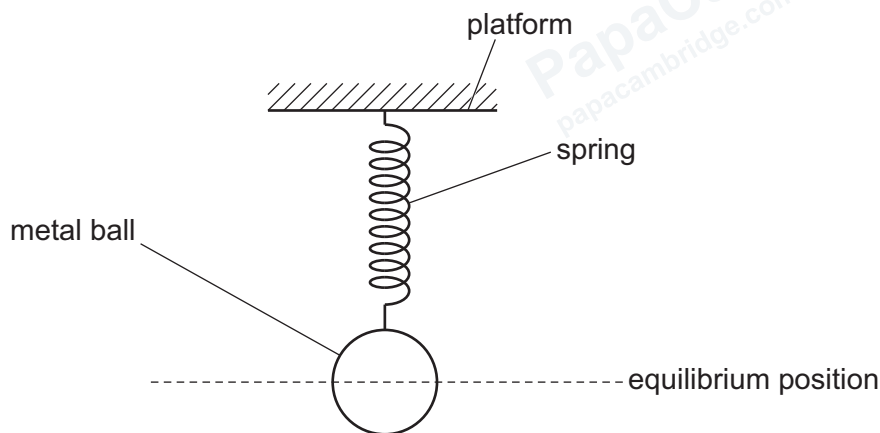


Figure 5.1

The upper end of the spring is attached to a platform that is initially stationary.

- (a) The ball is pulled down by a small distance and released so that it oscillates with simple harmonic motion. The oscillations have an amplitude of 2.5 cm and a maximum speed of 4.4 cm s^{-1} .
- (i) On Figure 5.2, sketch the variation of the velocity v of the ball with the displacement x from its equilibrium position as the ball undergoes one full oscillation.

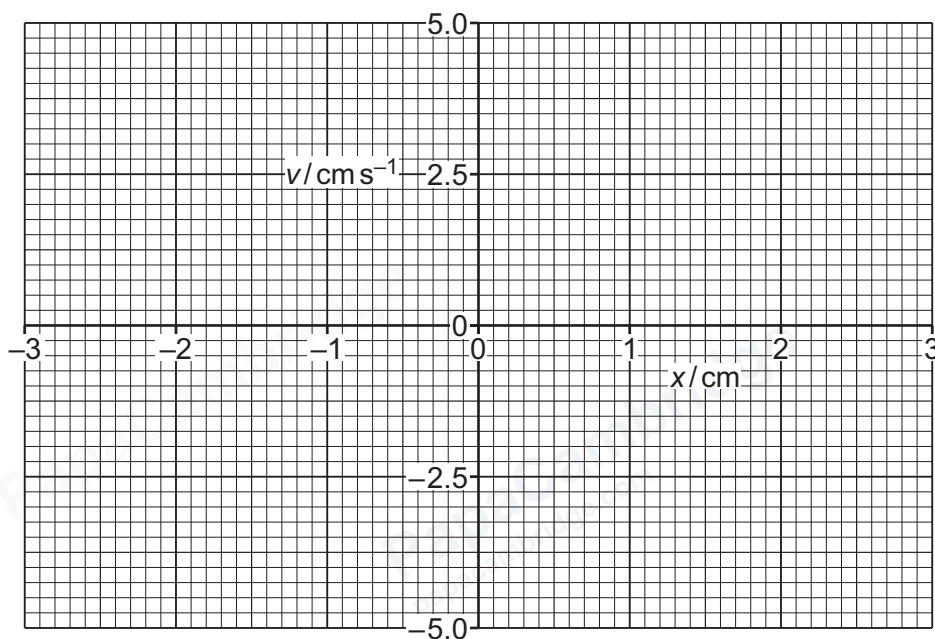


Figure 5.2

(ii) Show that the frequency of the oscillations is 0.28 Hz.

[2]

(b) The ball is returned to rest at its equilibrium position.

The platform is now moved up and down by a motor so that it oscillates vertically with simple harmonic motion of constant amplitude but variable frequency. The oscillating platform causes the ball to oscillate.

(i) State what is meant by resonance.

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(ii) The frequency f of the oscillations of the platform slowly increases from 0 to 0.40 Hz.

On Figure 5.3, sketch the variation with f of the amplitude of the oscillations of the **ball**.

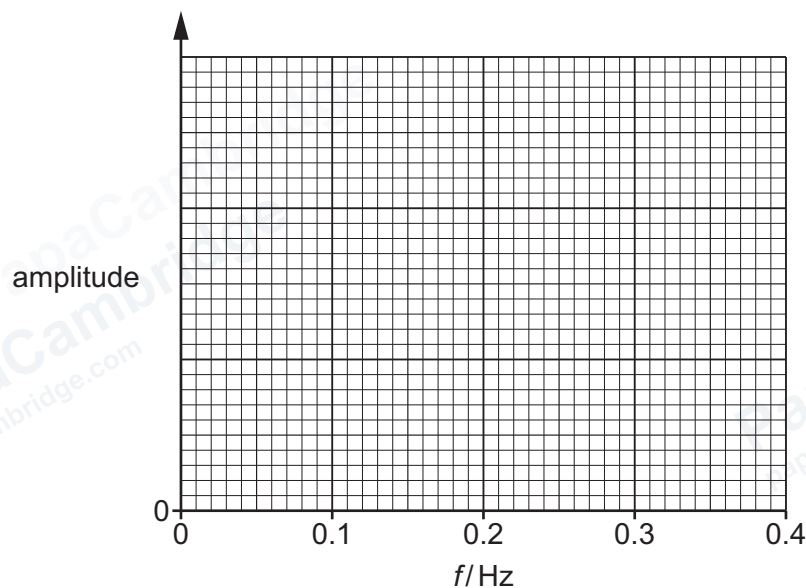


Figure 5.3

[2]

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